

A Rational Counterexample to the Four-Matricization Criterion in Symmetric Dimension Four

Candidate full counterexample packet
arXiv:2108.10405, Conjecture 1

11 August 2026

Status. Candidate full counterexample to Conjecture 1 of Champagne–Johnston–MacDonald–Pipes [1], subject to expert review. All certificates below are exact integers.

1 Source question

Champagne, Johnston, MacDonald, and Pipes characterize absolute symmetric PPT spectra using symmetric matricizations. Their Conjecture 1 asks whether, for an ordered spectrum $\lambda_1 \geq \dots \geq \lambda_{10} \geq 0$ on $\mathbb{C}^4 \vee \mathbb{C}^4$, it suffices to check just the four displayed matrices on PDF page 14 [1]. They report that a few million random examples supported the conjecture, while a proof remained elusive. The later lane-source paper of Xiong and Sze again describes the symmetric $m \geq 4$ case as open [2, p. 2].

Conjecture 1. Suppose $\rho \in \mathcal{L}(\mathbb{C}^4 \vee \mathbb{C}^4)$ has eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_{10} \geq 0$. Then ρ is absolutely symmetric PPT if and only if the following four matrices are positive semidefinite:

$$\begin{bmatrix} 2\lambda_{10} & \lambda_3 & \lambda_2 & \lambda_1 \\ \lambda_3 & 2\lambda_4 & \lambda_5 & \lambda_6 \\ \lambda_2 & \lambda_5 & 2\lambda_7 & \lambda_8 \\ \lambda_1 & \lambda_6 & \lambda_8 & 2\lambda_9 \end{bmatrix}, \quad \begin{bmatrix} 2\lambda_{10} & \lambda_3 & \lambda_2 & \lambda_1 \\ \lambda_3 & 2\lambda_4 & \lambda_5 & \lambda_7 \\ \lambda_2 & \lambda_5 & 2\lambda_6 & \lambda_8 \\ \lambda_1 & \lambda_7 & \lambda_8 & 2\lambda_9 \end{bmatrix},$$

$$\begin{bmatrix} 2\lambda_{10} & \lambda_6 & \lambda_2 & \lambda_1 \\ \lambda_6 & 2\lambda_5 & \lambda_4 & \lambda_3 \\ \lambda_2 & \lambda_4 & 2\lambda_7 & \lambda_8 \\ \lambda_1 & \lambda_3 & \lambda_8 & 2\lambda_9 \end{bmatrix}, \quad \begin{bmatrix} 2\lambda_{10} & \lambda_7 & \lambda_2 & \lambda_1 \\ \lambda_7 & 2\lambda_5 & \lambda_4 & \lambda_3 \\ \lambda_2 & \lambda_4 & 2\lambda_6 & \lambda_8 \\ \lambda_1 & \lambda_3 & \lambda_8 & 2\lambda_9 \end{bmatrix}.$$

Indeed, we have shown numerically that there does not exist a set of just three 4×4 symmetric matricizations whose positive semidefiniteness implies absolutely symmetric PPT of the associated quantum state, and a few million random examples suggest that the four matrices of Conjecture 1 suffice. However, a proof remains elusive (and a proof along the lines of the proof of Corollary 3 would be extremely long and tedious).

Figure 1: Conjecture 1 and its numerical motivation in the source PDF, page 14.

Transcription. If $\rho \in \mathcal{L}(\mathbb{C}^4 \vee \mathbb{C}^4)$ has eigenvalues $\lambda_1 \geq \dots \geq \lambda_{10} \geq 0$, the conjecture asserts that ρ is absolutely symmetric PPT if and only if the following four matrices are positive semidefinite:

$$\begin{aligned} M_6 &= \begin{bmatrix} 2\lambda_{10} & \lambda_3 & \lambda_2 & \lambda_1 \\ \lambda_3 & 2\lambda_4 & \lambda_5 & \lambda_6 \\ \lambda_2 & \lambda_5 & 2\lambda_7 & \lambda_8 \\ \lambda_1 & \lambda_6 & \lambda_8 & 2\lambda_9 \end{bmatrix}, & M_{13} &= \begin{bmatrix} 2\lambda_{10} & \lambda_3 & \lambda_2 & \lambda_1 \\ \lambda_3 & 2\lambda_4 & \lambda_5 & \lambda_7 \\ \lambda_2 & \lambda_5 & 2\lambda_6 & \lambda_8 \\ \lambda_1 & \lambda_7 & \lambda_8 & 2\lambda_9 \end{bmatrix}, \\ M_{11} &= \begin{bmatrix} 2\lambda_{10} & \lambda_6 & \lambda_2 & \lambda_1 \\ \lambda_6 & 2\lambda_5 & \lambda_4 & \lambda_3 \\ \lambda_2 & \lambda_4 & 2\lambda_7 & \lambda_8 \\ \lambda_1 & \lambda_3 & \lambda_8 & 2\lambda_9 \end{bmatrix}, & M_{12} &= \begin{bmatrix} 2\lambda_{10} & \lambda_7 & \lambda_2 & \lambda_1 \\ \lambda_7 & 2\lambda_5 & \lambda_4 & \lambda_3 \\ \lambda_2 & \lambda_4 & 2\lambda_6 & \lambda_8 \\ \lambda_1 & \lambda_3 & \lambda_8 & 2\lambda_9 \end{bmatrix}. \end{aligned}$$

2 Definitions and source reduction

For $x \in \mathbb{R}^{10}$, an upper-triangular matricization places the ten coordinates of x , in any order, in the upper triangle of a 4×4 matrix X . The corresponding *symmetric matricization* is $X + X^T$; thus diagonal entries are doubled and off-diagonal entries are not. Theorem 1 of [1] states that a symmetric state is absolutely symmetric PPT if and only if every symmetric matricization of its spectrum is positive semidefinite. Consequently, to disprove the reverse implication of the conjecture it is enough to make its four matrices positive semidefinite while one other symmetric matricization is indefinite.

3 Proof intuition

The four proposed tests differ only by interchanging $\lambda_5, \lambda_6, \lambda_7$ in a few slots. Making those three eigenvalues equal collapses the four tests to two. A second pair of repeated eigenvalues, $\lambda_8 = \lambda_9$, leaves enough freedom to keep those two matrices barely positive definite while changing the placement of λ_8 and λ_9 in another matricization produces a negative determinant. Strict inequalities on both sides make the failure robust, not a boundary artifact.

4 Exact counterexample

Theorem 1. *Let*

$$(\lambda_1, \dots, \lambda_{10}) = \frac{1}{8023}(1000, 1000, 1000, 1000, 703, 703, 703, 660, 660, 594). \quad (1)$$

Then $M_6, M_{13}, M_{11}, M_{12}$ are positive definite, but a symmetric matricization of this spectrum is not positive semidefinite. Hence any symmetric quantum state with spectrum (1) is not absolutely symmetric PPT, and Conjecture 1 of [1] is false.

Proof. The ten displayed integers are nonincreasing, positive, and sum to 8023, so (1) is an ordered density-matrix spectrum. Multiplying all matricizations by the positive number 8023 does not affect definiteness. The equalities among the weights give

$$8023M_6 = 8023M_{13} = A := \begin{bmatrix} 1188 & 1000 & 1000 & 1000 \\ 1000 & 2000 & 703 & 703 \\ 1000 & 703 & 1406 & 660 \\ 1000 & 703 & 660 & 1320 \end{bmatrix}$$

and

$$8023M_{11} = 8023M_{12} = B := \begin{bmatrix} 1188 & 703 & 1000 & 1000 \\ 703 & 1406 & 1000 & 1000 \\ 1000 & 1000 & 1406 & 660 \\ 1000 & 1000 & 660 & 1320 \end{bmatrix}.$$

Their leading principal minors, in orders 1, 2, 3, 4, are respectively

$$\begin{aligned} \Delta(A) &= (1188, 1,376,000, 753,535,708, 293,705,189,448), \\ \Delta(B) &= (1188, 1,176,119, 465,623,314, 137,338,080). \end{aligned}$$

Every entry in both lists is positive, so Sylvester's criterion makes A and B positive definite.

Now place the spectrum entries in the upper triangle, read row by row, in the order

$$(10, 9, 3, 1, 7, 4, 2, 5, 6, 8).$$

(This is also ordering 4 in the source authors’ list of 26 relevant orders.) The resulting symmetric matricization, again scaled by 8023, is

$$C := \begin{bmatrix} 1188 & 660 & 1000 & 1000 \\ 660 & 1406 & 1000 & 1000 \\ 1000 & 1000 & 1406 & 703 \\ 1000 & 1000 & 703 & 1320 \end{bmatrix}.$$

Direct exact elimination gives

$$\det C = -337,300,392 < 0.$$

A positive-semidefinite matrix has nonnegative determinant, so $C/8023$ is not positive semidefinite. Theorem 1 of [1] therefore shows that a state with spectrum (1) is not absolutely symmetric PPT, despite passing all four proposed tests. Such a state exists, for example as the diagonal operator with these eigenvalues in any orthonormal basis of $\mathbb{C}^4 \vee \mathbb{C}^4$. $\square$

5 Independent verification

The included script `code/check_counterexample.py` uses only exact integer arithmetic. It verifies the order and normalization of (1), constructs the four matrices directly from the source formulas, confirms the two pairwise equalities and all eight leading minors, constructs C from its placement order, and confirms its negative determinant. It implements fraction-free Bareiss elimination rather than floating-point eigensolvers. Running it prints:

```
M6=M13 leading principal minors: (1188, 1376000, 753535708, 293705189448)
M11=M12 leading principal minors: (1188, 1176119, 465623314, 137338080)
ordering-4 determinant: -337300392
all exact checks passed
```

The construction was discovered by constrained semidefinite numerical search, then rationalized and rechecked independently in exact arithmetic. Two failed proof routes—coefficientwise domination of principal-minor polynomials and direct Loewner domination by the four tests—were also checked before the counterexample search succeeded.

6 Scope, novelty, and review recommendation

This packet refutes exactly the four-matrix sufficiency assertion. It does not identify a minimal replacement subfamily among the 26 relevant matricizations, and it does not decide whether absolutely symmetric separable and absolutely symmetric PPT states coincide in dimension four.

A bounded novelty search through 11 August 2026 covered the run’s four cheap indexes; the exact arXiv id and title; the phrases “four matrices”, “absolutely symmetric PPT”, “Conjecture 1”, “symmetric matricization”, and “counterexample”; the current arXiv record; and the 2022 journal metadata. It recovered the source conjecture and no later proof or counterexample. This is not a claim of priority.

Human-review recommendation. High priority. Check the transcription of the four source matrices and the upper-triangular placement defining C . After that, the certificate consists only of the displayed integer minors and determinant and is readily auditable.

References

- [1] G. Champagne, N. Johnston, M. MacDonald, and L. Pipes, *Spectral Properties of Symmetric Quantum States and Symmetric Entanglement Witnesses*, Linear Algebra Appl. 649 (2022), 273–300; [arXiv:2108.10405](#).
- [2] L. Xiong and N.-S. Sze, *Criteria of absolutely separability from spectrum for qudit-qudits states*, [arXiv:2408.11684](#) (2024).